

Math 131A: Analysis

Discussion 10: Differentiation and Integrability

1 Differentiation

Definition: Let $I = (a, b) \subset \mathbb{R}$, $c \in I$. If the following limit exists and is finite, then the *derivative* of $f : I \rightarrow \mathbb{R}$ at c is

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}.$$

f is *differentiable* on I if $f'(c)$ exists for all $c \in I$.

Mean Value Theorem: Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists a $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

1. (#6, 28.2) Calculate the derivatives of the following functions at the indicated points

(i) $f(x) = x^3$ at $x = 2$

(ii) $f(x) = x + 2$ at $x = a$

(iii) $f(x) = x^2 \cos x$ at $x = 0$

(iv) $f(x) = \frac{3x+4}{2x-1}$ at $x = 1$

2. (#7) Suppose f is differentiable on $\mathbb{R}$ and $f(0) = 0$, $f(1) = 1$, and $f(2) = 1$.

(a) Show that $f'(x) = \frac{1}{2}$ for some $x \in (0, 2)$.

(b) Show that $f'(x) = \frac{1}{7}$ for some $x \in (0, 2)$.

3. (#10, 29.5) Let f be defined on $\mathbb{R}$, and suppose $|f(x) - f(y)| \leq (x - y)^2$ for all $x, y \in \mathbb{R}$. Prove f is a constant function.

4. (#12, 29.13) Prove that if f, g are differentiable on $\mathbb{R}$, $f(0) = g(0)$, and $f'(x) \leq g'(x)$ for all $x \in \mathbb{R}$, then $f(x) \leq g(x) \forall x \geq 0$.

2 Riemann Integrability

Definition: The *upper Darboux sum* $U(f, P)$ of f with respect to partition P is

$$U(f, P) = \sum_{t=1}^n \max(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}).$$

The *lower Darboux sum* $L(f, P)$ of f with respect to partition P is

$$L(f, P) = \sum_{t=1}^n \min(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}).$$

f is *Riemann integrable* if

$$U(f) = \inf\{U(f, P) : P \text{ a partition of } [a, b]\} = \sup\{L(f, P) : P \text{ a partition of } [a, b]\} = L(f)$$

1. Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$.
 - (a) Show that if one value of $f(x)$ is changed at some point $x \in [a, b]$, this does not affect the integral and integrability of f .
 - (b) Show this remains true if we change a finite number of values
 - (c) Find an example to show that altering f at an infinite number of points may cause the resulting function to no longer be Riemann integrable.
2. Is it possible to have a function $f : [a, b] \rightarrow \mathbb{R}$ with infinitely many discontinuities that is Riemann integrable?